

Database Schema Document

Rwanda NBSAP Monitoring Dashboard 2025–2030

Prepared for: Rwanda Environment Management Authority (REMA) **Document Reference:** NBSAP-DSD-2026-006 **Version:** 1.0 **Date:** June 2026 **Classification:** Official — Government Use

Document Control

Field	Value
Document Title	Database Schema Document
Project	Rwanda NBSAP Monitoring Dashboard
Database Engine	PostgreSQL 15 (Supabase-managed)
Prepared by	NBSAP Dashboard Development Team
Approved by	REMA — Department of Biodiversity & Landscape Management
Status	Final

1. Overview

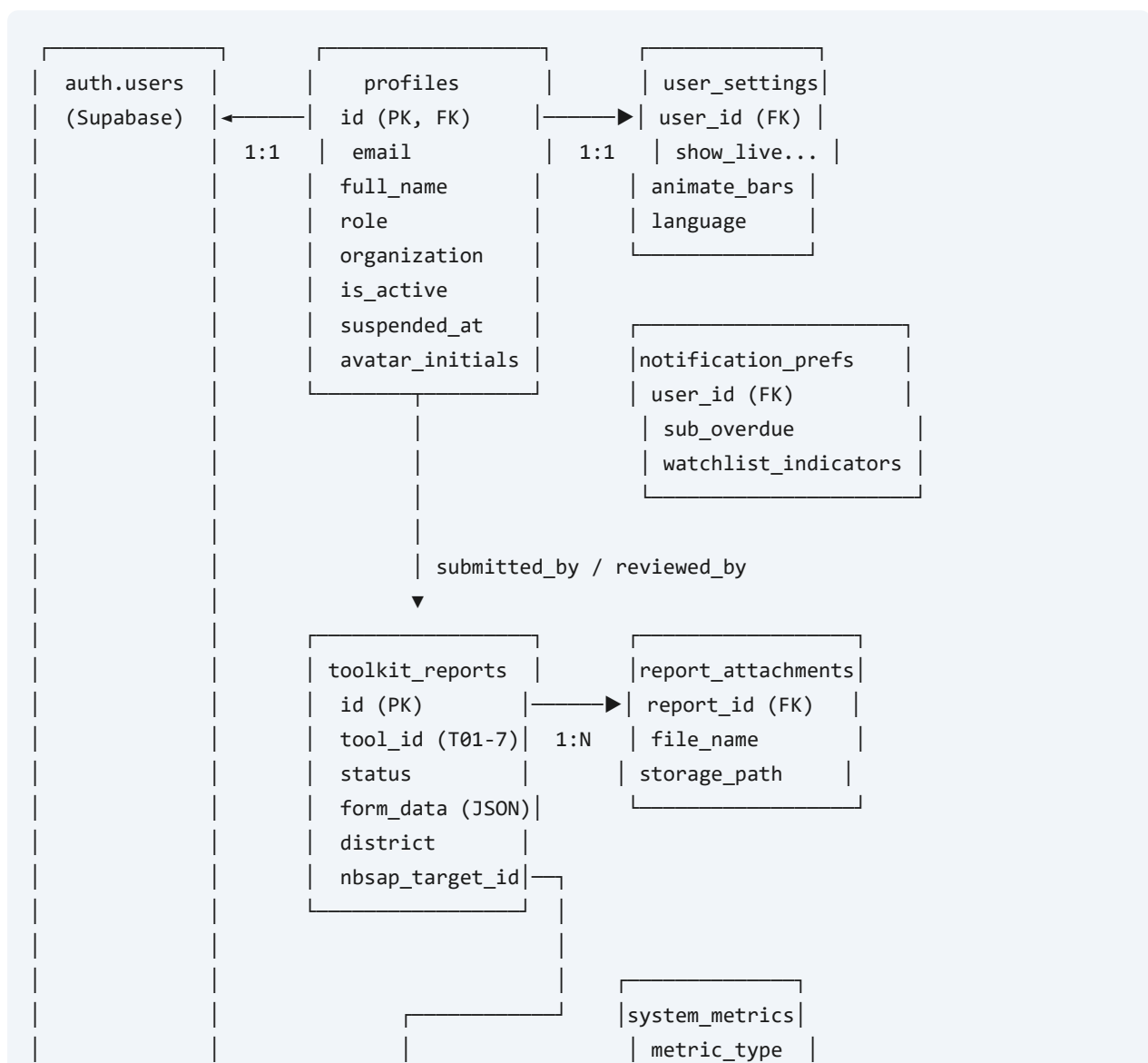
1.1 Database Summary

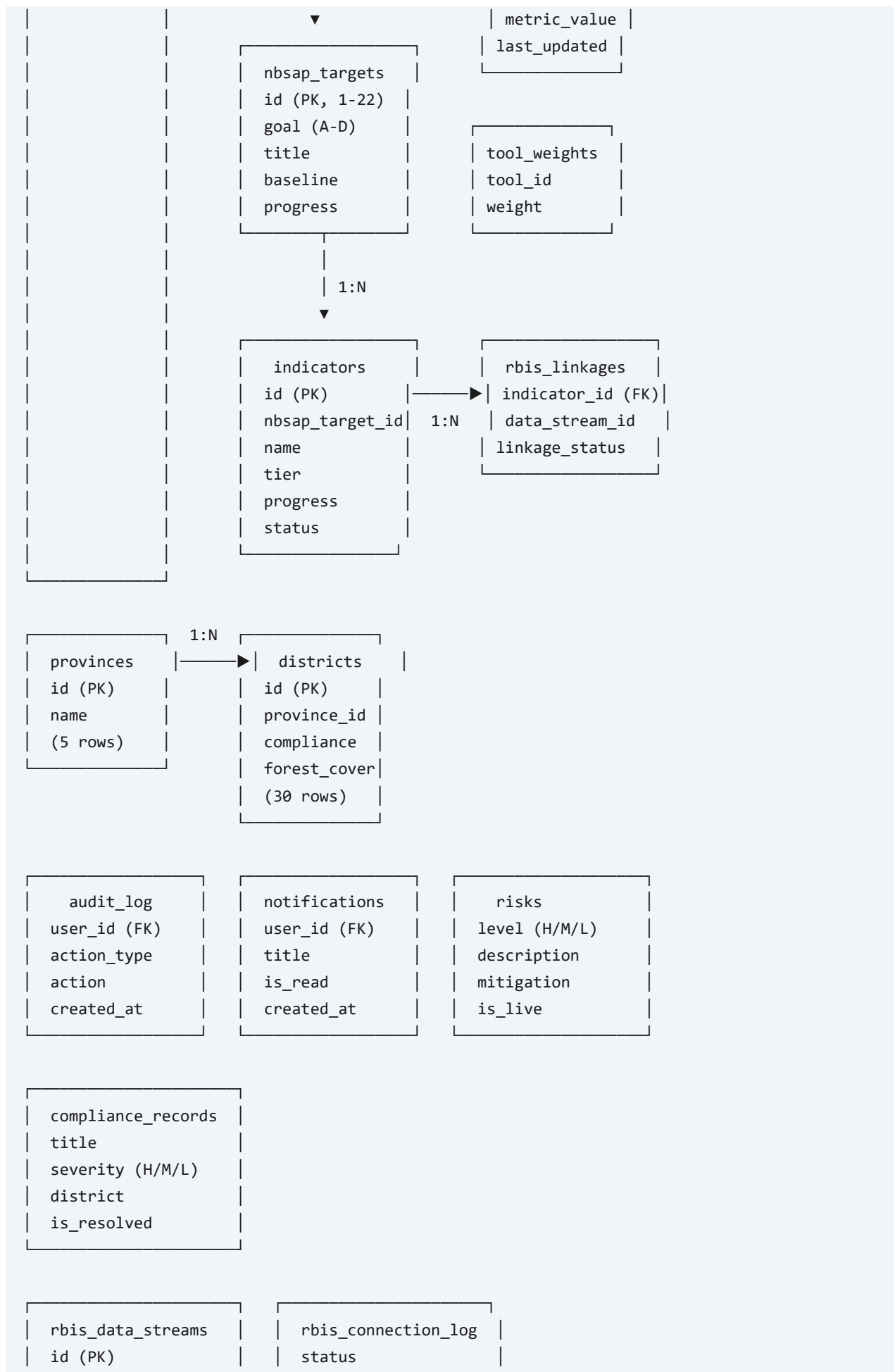
Attribute	Value
Engine	PostgreSQL 15
Hosting	Supabase Cloud (managed)
Total Tables	18
Total Indexes	20+
RLS Policies	25+
Triggers	7
Functions	10+
Migrations	22 (numbered 001–022)

1.2 Schema Categories

Category	Tables	Purpose
User Management	4	User accounts, settings, notification preferences, role change requests
Core Biodiversity Data	3	NBSAP targets, indicators, milestones
Reporting	2	Toolkit report submissions, file attachments
Metrics & Weights	2	Automated system metrics, tool weight configuration
Governance	4	Compliance records, risk register, audit log, notifications
Geography	3	Provinces, districts, lakes
RBIS Integration	3	RBIS linkages, data streams, connection log

2. Entity-Relationship Diagram





name	server_url
target_numbers[]	user_id (FK)
occurrence_count	created_at
status	

3. Table Specifications

3.1 User Management Tables

3.1.1 profiles

Extends Supabase `auth.users` with application-specific fields.

Column	Type	Constraints	Description
id	UUID	PK, FK → auth.users	User unique identifier
email	TEXT	NOT NULL	User email address
full_name	TEXT		User full name
role	user_role (ENUM)	NOT NULL, DEFAULT 'policy_monitoring'	One of 6 system roles
organization	TEXT		User's organisation
department	TEXT		User's department
phone	TEXT		Contact phone number
is_active	BOOLEAN	DEFAULT TRUE	Account active status
suspended_at	TIMESTAMPTZ		When account was suspended
suspended_by	UUID	FK → profiles	Admin who suspended
suspension_reason	TEXT		Reason for suspension
suspension_end_date	TIMESTAMPTZ		Auto-reactivation date
last_login	TIMESTAMPTZ		Last login timestamp
avatar_initials	TEXT	GENERATED (first 2 chars of name)	UI display initials
created_at	TIMESTAMPTZ	DEFAULT NOW()	Account creation time
updated_at	TIMESTAMPTZ	DEFAULT NOW()	Last profile update

3.1.2 user_settings

Per-user dashboard preferences.

Column	Type	Default	Description
user_id	UUID	PK, FK → profiles	User reference
show_live_stats	BOOLEAN	TRUE	Display live statistics
animate_bars	BOOLEAN	TRUE	Animate progress bars
compact_sidebar	BOOLEAN	FALSE	Compact sidebar layout
auto_refresh	BOOLEAN	TRUE	Auto-refresh dashboard data
show_baseline	BOOLEAN	TRUE	Show baseline values
require_verification	BOOLEAN	TRUE	Require data verification
language	TEXT	'en'	Interface language
updated_at	TIMESTAMPZ	NOW()	Last settings update

3.1.3 notification_preferences

Per-user notification subscription settings.

Column	Type	Default	Description
user_id	UUID	PK, FK → profiles	User reference
email	TEXT		Email for notifications
phone	TEXT		Phone for notifications
sub_overdue	BOOLEAN	TRUE	Subscribe to overdue alerts
sub_compliance	BOOLEAN	TRUE	Subscribe to compliance alerts
compliance_threshold	INTEGER	60	Compliance alert threshold (%)
sub_deadlines	BOOLEAN	TRUE	Subscribe to deadline reminders
deadline_days	INTEGER	14	Days before deadline to alert
sub_pending	BOOLEAN	TRUE	Subscribe to pending report alerts
sub_finance	BOOLEAN	TRUE	Subscribe to finance alerts
sub_risk	BOOLEAN	TRUE	Subscribe to risk alerts
watchlist_indicators	INTEGER[]		Array of indicator IDs to watch

3.1.4 role_change_requests

Formal workflow for user role change requests.

Column	Type	Description
id	UUID	PK
user_id	UUID	FK → profiles (requester)
current_role	user_role	Role at time of request
requested_role	user_role	Desired new role
justification	TEXT	Reason for request
status	TEXT	'pending' / 'approved' / 'rejected'
reviewed_by	UUID	FK → profiles (approver/rejecter)
review_note	TEXT	Reviewer's comment
created_at	TIMESTAMPTZ	Request timestamp
reviewed_at	TIMESTAMPTZ	Review timestamp

3.2 Core Biodiversity Data Tables

3.2.1 nbsap_targets

Rwanda's 22 NBSAP 2025–2030 targets.

Column	Type	Constraints	Description
id	INTEGER	PK (1–22)	Target number
goal	gbf_goal (ENUM)	NOT NULL	KM-GBF Goal: A, B, C, or D
title	TEXT	NOT NULL	Official target title
description	TEXT		Full target statement
baseline	TEXT		Baseline data description
timeline_milestones	TEXT		Phase-by-phase milestones
progress	INTEGER	CHECK 0–100, DEFAULT 0	Auto-calculated progress (%)
created_at	TIMESTAMPTZ	DEFAULT NOW()	Record creation
updated_at	TIMESTAMPTZ	DEFAULT NOW()	Last update

Seed Data: 22 targets across 4 goals (Goal A: 1–4, Goal B: 5–8, Goal C: 9–12, Goal D: 13–22)

3.2.2 indicators

~80 biodiversity indicators mapped to NBSAP targets.

Column	Type	Constraints	Description
id	SERIAL	PK	Auto-incrementing ID
name	TEXT	NOT NULL	Indicator name
definition	TEXT		Detailed definition
tier	indicator_tier (ENUM)	NOT NULL	headline / component / complementary / binary
nbsap_target_id	INTEGER	FK → nbsap_targets	Parent target
target_2030	TEXT		2030 target value
baseline	TEXT		Baseline value
midterm	TEXT		Midterm target value
final_target	TEXT		Final target value
current_value	TEXT		Current measured value
progress	INTEGER	CHECK 0–100, DEFAULT 0	Progress percentage
status	indicator_status (ENUM)	DEFAULT 'behind'	on-track / at-risk / behind
km_gbf	TEXT		KM-GBF indicator reference
periodicity	TEXT		Measurement frequency
data_source	TEXT		Data source description
responsible	TEXT[]		Array of responsible institutions
created_at	TIMESTAMPTZ	DEFAULT NOW()	Record creation
updated_at	TIMESTAMPTZ	DEFAULT NOW()	Last update

Seed Data: ~79 indicators distributed across 22 targets in 4 tiers

3.2.3 nbsap_milestones

Implementation milestones for each target, 2025–2030.

Column	Type	Description
id	SERIAL	PK

target_id	INTEGER	FK → nbsap_targets
phase	TEXT	Implementation phase (e.g., "2025-2026")
milestone	TEXT	Milestone description
status	TEXT	Current status

3.3 Reporting Tables

3.3.1 toolkit_reports

All T01–T07 report submissions.

Column	Type	Constraints	Description
id	UUID	PK (auto-generated)	Report unique identifier
tool_id	report_type (ENUM)	NOT NULL	T01–T07
tool_name	TEXT	NOT NULL	Human-readable tool name
submitted_by	UUID	FK → profiles	Reporter
status	submission_status (ENUM)	DEFAULT 'pending'	pending / approved / rejected
reviewed_by	UUID	FK → profiles	Reviewer (null until reviewed)
reviewed_at	TIMESTAMPTZ		Review timestamp
review_note	TEXT		Reviewer's note
period	TEXT		Reporting period
form_data	JSONB	NOT NULL, DEFAULT '{}'	Dynamic form fields
attachments	JSONB	DEFAULT '[]'	Attachment metadata
district	TEXT		District (denormalised)
institution	TEXT		Institution (denormalised)
nbsap_target_id	INTEGER	FK → nbsap_targets	Linked target
submitted_at	TIMESTAMPTZ	DEFAULT NOW()	Submission time
created_at	TIMESTAMPTZ	DEFAULT NOW()	Record creation
updated_at	TIMESTAMPTZ	DEFAULT NOW()	Last update

JSONB form_data **Examples by Tool:**

Tool	Key Fields in form_data
T01	institution , budget_utilised , compliance_score , policy_progress
T02	district , forest_hectares , wetland_hectares , land_cover_change
T03	area_name , coverage_km2 , management_effectiveness , illegal_cases , restoration_ha
T04	incident_type , district , species , hwc_case_count , mitigation_measures
T05	budget_allocated , budget_disbursed , finance_source
T06	company , eia_status , sector , restoration_commitments
T07	community_name , district , participant_count , traditional_knowledge

3.3.2 report_attachments

Evidence files linked to reports.

Column	Type	Description
id	UUID	PK
report_id	UUID	FK → toolkit_reports (CASCADE DELETE)
file_name	TEXT	Original filename
file_ext	TEXT	File extension
file_size	BIGINT	File size in bytes
storage_path	TEXT	Supabase Storage path
uploaded_by	UUID	FK → profiles
created_at	TIMESTAMPTZ	Upload timestamp

3.4 Metrics & Weights Tables

3.4.1 system_metrics

13 automated system-wide metrics computed from approved reports.

Column	Type	Description
id	SERIAL	PK
metric_type	TEXT	Metric identifier
metric_value	NUMERIC	Current computed value

metric_label	TEXT	Human-readable label
unit	TEXT	Unit of measurement
last_updated	TIMESTAMPTZ	Last recalculation time

13 Metric Types:

Metric Type	Label	Unit	Source Tool
forest_restored_ha	Forest Area Restored	Hectares	T02
wetland_restored_ha	Wetland Area Restored	Hectares	T02
protected_area_coverage	Protected Area Coverage	%	T03
mgmt_effectiveness_avg	Avg Management Effectiveness	Score	T03
illegal_cases_total	Total Illegal Cases Reported	Count	T03
hwc_incidents_total	Total HWC Incidents	Count	T04
budget_allocated_total	Total Budget Allocated	RWF	T05
budget_disbursed_total	Total Budget Disbursed	RWF	T05
disbursement_rate	Budget Disbursement Rate	%	T05
eia_compliance_rate	EIA Compliance Rate	%	T06
restoration_commitments	Private Sector Restoration	Hectares	T06
community_participants	Community Participants	Count	T07
traditional_knowledge_entries	Traditional Knowledge Entries	Count	T07

3.4.2 tool_weights

Configurable weights for each reporting tool's contribution to target progress.

Column	Type	Description
id	SERIAL	PK
tool_id	report_type	T01–T07
target_id	INTEGER	FK → nbsap_targets
weight	NUMERIC	Weight value (0.0–1.0)
updated_at	TIMESTAMPTZ	Last weight change

3.5 Governance Tables

3.5.1 audit_log

Immutable, append-only log of all system actions.

Column	Type	Description
id	UUID	PK
user_id	UUID	FK → profiles (actor)
action_type	TEXT	submit / approve / reject / export / view / login / delete
action	TEXT	Human-readable action description
detail	TEXT	Additional context
role	TEXT	User's role at time of action
ip_address	TEXT	Client IP address
created_at	TIMESTAMPTZ	Action timestamp

3.5.2 risks

Biodiversity risk register.

Column	Type	Description
id	TEXT	PK (risk identifier)
description	TEXT	Risk description
category	TEXT	Risk category
likelihood	TEXT	Likelihood assessment
impact	TEXT	Impact assessment
level	risk_level (ENUM)	High / Medium / Low
mitigation	TEXT	Mitigation strategy
owner	TEXT	Risk owner
is_live	BOOLEAN	Whether auto-generated from submissions

3.5.3 compliance_records

Column	Type	Description
id	UUID	PK
title	TEXT	Compliance item title

description	TEXT	Details
severity	TEXT	High / Medium / Low
district	TEXT	Affected district
flagged_at	TIMESTAMPTZ	When flagged
resolved_at	TIMESTAMPTZ	When resolved
resolved_by	UUID	FK → profiles
is_resolved	BOOLEAN	Resolution status

3.5.4 notifications

Column	Type	Description
id	UUID	PK
user_id	UUID	FK → profiles (recipient)
title	TEXT	Notification title
message	TEXT	Notification body
type	TEXT	info / warning / error / success
is_read	BOOLEAN	Read status
action_tab	TEXT	Navigation target
action_label	TEXT	Action button label
created_at	TIMESTAMPTZ	Notification timestamp

3.6 Geography Tables

3.6.1 provinces

Rwanda's 5 provinces. **Data:** Kigali, South, North, West, East

Column	Type	Description
id	SERIAL	PK
name	TEXT	Province name (unique)

3.6.2 districts

Rwanda's 30 districts with compliance tracking.

Column	Type	Description
id	SERIAL	PK
name	TEXT	District name (unique)
province_id	INTEGER	FK → provinces
status	district_status (ENUM)	submitted / pending / missing
compliance	INTEGER	Compliance score (0–100)
forest_cover	INTEGER	Forest cover percentage (0–100)
latitude	NUMERIC	GPS latitude
longitude	NUMERIC	GPS longitude
elevation	NUMERIC	Elevation (metres)
wetland_area	NUMERIC	Wetland area

3.6.3 lakes

Rwanda's inland water bodies.

Column	Type	Description
id	SERIAL	PK
name	TEXT	Lake name
province	TEXT	Province location
area_km2	NUMERIC	Surface area
latitude	NUMERIC	GPS latitude
longitude	NUMERIC	GPS longitude

3.7 RBIS Integration Tables

3.7.1 rbis_linkages

Column	Type	Description
id	SERIAL	PK
indicator_id	INTEGER	FK → indicators
data_stream_id	VARCHAR(100)	RBIS data stream identifier

linkage_status	VARCHAR(20)	linked / not-linked / partial
last_sync	TIMESTAMPTZ	Last synchronisation time

3.7.2 rbis_data_streams

8 predefined data streams: Protected Areas Coverage, Threatened Species Monitoring, Forest Cover Change, Wetland Extent, Species Distribution, Invasive Species Tracking, Ecosystem Restoration, Sustainable Use Indicators.

Column	Type	Description
id	VARCHAR(100)	PK
name	VARCHAR(255)	Display name
description	TEXT	Stream description
target_numbers	INTEGER[]	Array of linked target IDs
occurrence_count	INTEGER	Current occurrence count
status	VARCHAR(20)	active / inactive / error
icon	VARCHAR(50)	UI icon identifier
color	VARCHAR(20)	UI colour code

3.7.3 rbis_connection_log

Column	Type	Description
id	SERIAL	PK
status	VARCHAR(20)	connected / disconnected / error
server_url	VARCHAR(255)	RBIS server URL
error_message	TEXT	Error details
user_id	UUID	FK → auth.users
created_at	TIMESTAMPTZ	Event timestamp

4. Enumerated Types

Type Name	Values	Usage
user_role	policy_monitoring, lead_government_ministry_reporting, local_reporting, dashboard_management,	User access level

	programme_alignment, public_viewer	
indicator_status	on-track, at-risk, behind	Indicator progress status
indicator_tier	headline, component, complementary, binary	KM-GBF indicator tier
submission_status	pending, approved, rejected	Report verification status
district_status	submitted, pending, missing	District reporting status
report_type	T01, T02, T03, T04, T05, T06, T07	Reporting tool identifier
risk_level	High, Medium, Low	Risk severity
gbf_goal	A, B, C, D	KM-GBF Goal identifier

5. Index Inventory

Index Name	Table	Columns	Purpose
idx_toolkit_reports_tool_id	toolkit_reports	tool_id	Filter reports by tool
idx_toolkit_reports_submitted_by	toolkit_reports	submitted_by	Find reports by user
idx_toolkit_reports_status	toolkit_reports	status	Filter by verification status
idx_toolkit_reports_submitted_at	toolkit_reports	submitted_at DESC	Sort by submission date
idx_indicators_target	indicators	nbsap_target_id	Find indicators by target
idx_indicators_tier	indicators	tier	Filter by indicator tier
idx_indicators_status	indicators	status	Filter by indicator status
idx_audit_log_user	audit_log	user_id	Find actions by user
idx_audit_log_created	audit_log	created_at DESC	Sort by timestamp

<code>idx_notifications_user</code>	notifications	user_id, is_read	Unread notifications query
<code>idx_rbis_linkages_indicator</code>	rbis_linkages	indicator_id	Find linkages by indicator
<code>idx_rbis_linkages_stream</code>	rbis_linkages	data_stream_id	Find linkages by stream
<code>idx_rbis_streams_status</code>	rbis_data_streams	status	Filter streams by status
<code>idx_rbis_connection_log_created</code>	rbis_connection_log	created_at DESC	Recent connection events

6. Row-Level Security (RLS)

All 18 tables have RLS enabled. Key policies:

Table	Policy	Rule
profiles	Users view own profile	<code>auth.uid() = id</code>
profiles	Admins view all profiles	<code>is_admin()</code> returns TRUE
profiles	Users update own profile	<code>auth.uid() = id</code>
profiles	Admins update any profile	<code>is_admin()</code> returns TRUE
indicators	All authenticated users read	<code>auth.uid() IS NOT NULL</code>
indicators	Writers can update	<code>can_write()</code> returns TRUE
indicators	Admins can insert	<code>is_admin()</code> returns TRUE
nbsap_targets	All authenticated users read	<code>auth.uid() IS NOT NULL</code>
toolkit_reports	Reporters read own + admins read all	Role-based SELECT
toolkit_reports	Writers can insert	<code>can_write()</code> returns TRUE
audit_log	Admins only	<code>is_admin()</code> returns TRUE
notifications	Users read own	<code>auth.uid() = user_id</code>

RLS Helper Functions

Function	Return Type	Purpose
<code>get_user_role()</code>	user_role	Returns current user's role
<code>is_role(check_role)</code>	BOOLEAN	Checks if user has specific role

<code>can_write()</code>	BOOLEAN	Checks if user has write permission
<code>is_admin()</code>	BOOLEAN	Checks if user is dashboard_management

7. Database Triggers

Trigger	Table	Event	Action
on_report_approved	toolkit_reports	UPDATE (status → 'approved')	Extract metrics from form_data; update system_metrics
on_report_deleted	toolkit_reports	DELETE (if approved)	Reverse metric contributions
on_weighted_progress	system_metrics	UPDATE	Recalculate target progress using tool_weights
on_profile_created	auth.users	INSERT	Create matching profiles row
on_role_change_approved	role_change_requests	UPDATE (status → 'approved')	Update user role in profiles
on_notification_created	notifications	INSERT	Push via Supabase Realtime
on_audit_action	Various	INSERT/UPDATE/DELETE	Append to audit_log

8. Database Functions

Function	Parameters	Returns	Purpose
<code>calculate_weighted_progress</code>	target_id INTEGER	NUMERIC	Compute target progress from tool-weighted metrics
<code>recalculate_all_metrics</code>	None	VOID	Full recalculation of all 13 system metrics
<code>extract_metric_from_report</code>	report_id UUID	JSONB	Extract metric-relevant values from report form_data
<code>reverse_metric_contribution</code>	report_id UUID	VOID	Undo metric impact of a deleted approved report
<code>get_user_role</code>	None (uses auth.uid())	user_role	Return current user's role

is_role	check_role user_role	BOOLEAN	Check user against a specific role
can_write	None	BOOLEAN	Check if current user has write access
is_admin	None	BOOLEAN	Check if current user is administrator
update_district_compliance	district_id INTEGER	VOID	Recalculate district compliance score

*Prepared by: NBSAP Dashboard Development Team Document Reference: NBSAP-DSD-2026-006
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